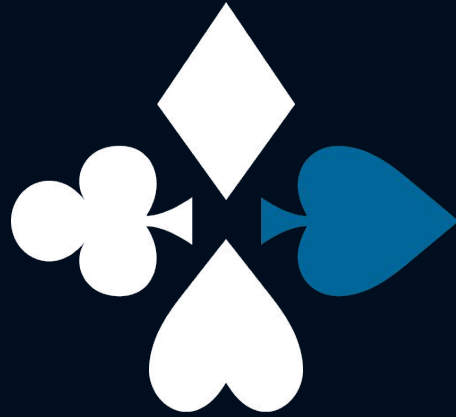




Pokerbots 2026

Lecture 5: Game Theory

Sponsors



CITADEL | CITADEL Securities





Announcements

GTO Wizard Challenge!

- Normal Heads-Up (Not Toss or Hold-em)
- First Place: \$2,500
- Second Place: \$1,500
- Third Place: \$1,000
- pkr.bot/gto



Week 1 Mini Tournament Results

WINNER [\$1000]:

spongebot

BIGGEST UPSET [\$500]:

don't panic

Tournament ELO 1156, won majority matches against “ling ling” (ELO 2146)

Week 1 Mini Tournament Results

Full tournament results
can be viewed at the
scrimmage server

Pokerbots Scrimmage

Home

Announcements

Manage Team

Your Games

Tournaments

Charts

Playground

Admin

Sponsor

Log out bcostin

Tournaments

Week 1 tournament

Private

2 days ago

75

PARTICIPANTS

50

GAMES/PAIR

Done!

Name	ELO	Wins	Losses
spongebot	2572	3583	117
Los Camellos	2424	3480	220
heyo999	2390	3430	257
Policy Gradient Is All You Need	2323	3338	358
prw	2313	3347	352
72o all-in	2306	3351	349
redacted	2234	3257	443
ling ling	2146	3110	590
confused	2132	3096	604
beepboop	2098	3052	647
no ragrets	2085	3034	665

staff-team-2

1564

1885

1815

yall gotta lock in

Make-up Progress Reports due Tuesday

- As per syllabus, if your team missed the Week 1 bot submission deadline, a progress report is required to earn credit
- Reports are half a page and must list team name and members
- Submit via email to pokerbots@mit.edu by 11:59pm Tuesday (Tomorrow)

Hackathon on Wednesday

- Wednesday night (1/14)
- 32-124 or 34-501, 6pm -- LATE
- Show up and grind
- Dinner provided
- Snacks, fun, and games
- Prizes for challenges and those that stick around
- Details will be announced on Piazza

Snack/Drink Recs?

[pkrfbot/snack](https://pkrfbot.com/snack)

No Class Thursday

Cancelled in light of Hackathon the night before

Recitation / Office Hours

- Recall new room: **4-257**



Giveaway Games!

First game: 2/3 Game ("Beauty Contest")

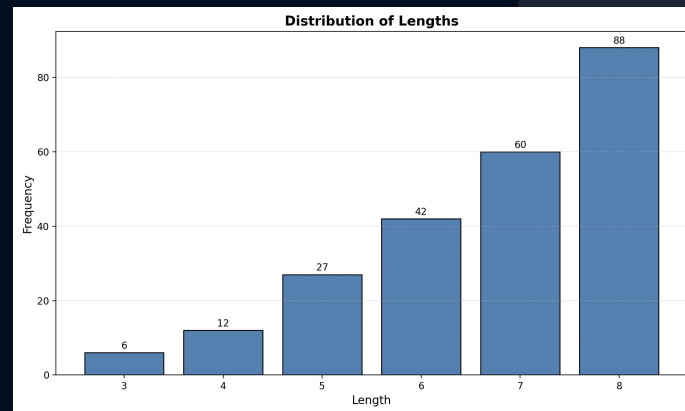
- Submit an integer between 0-1000 inclusive
- Winner is the closest guess to $\frac{2}{3}$ of the average submission (a tie is broken randomly)
 - i.e. if guesses are 100, 200, 300, 400, $\text{avg} = 250$, $\frac{2}{3} \text{ avg} = 166.67$, 200 wins
- Prize: Sony WF-1000XM5 Earbuds

pkr.bot/beauty



Second game: Kerb Bucket Wager

- 'Entry fee': your kerb gets blacklisted from winning giveaways for the next week
- Competitors bucketed by kerb length, smallest non-zero bucket wins (in case of ties, chosen at random)
- EVERYONE in winning bucket wins:
 - \$20



pkr.bot/lemons

Agenda

- What is a game?
- Pure and mixed strategies
- Nash equilibria
- Applications to poker



What is a game?

Definition

We generally only consider two-player zero-sum games; we will refer to these as simply *games*.

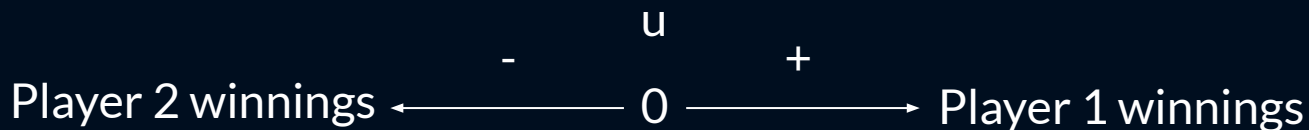
Examples: RPS, tic-tac-toe, (chess? poker??)

Formalized

A *game* between players 1 and 2 consists of a pair of strategy sets S_1 and S_2 , and a utility function $u : S_1 \times S_2 \rightarrow \mathbb{R}$. Players submit strategies simultaneously. Player 1 seeks to maximize u , and player 2 seeks to minimize u . If randomness is involved, then expected utility is used.

Think of utility as chips. This means players want to submit actions that have the highest expected payout.

Player 2 is minimizing since we're talking about zero-sum games.





Pure and mixed strategies

Pure and mixed strategies

- A mixed strategy is described by a probability distribution over pure strategies:
 - Example:
 - $P(\text{rock}) = 0.4$
 - $P(\text{paper}) = 0.3$
 - $P(\text{scissors}) = 0.3$
- There are only 3 pure strategies in RPS, but infinitely many mixed strategies
- Mixed strategies show us the power of randomness

Matrix form games

Suppose the strategy sets S_1 and S_2 consist of probability distributions over a finite list of pure strategies. Also, suppose that the utility function is linear

Linear = utility of a mixed strategy is expected utility over pure strategies

Then, we can write our game in *matrix form*, with utility function shown.

$S_1 \setminus S_2$	Rock	Paper	Scissors
Rock	0	-1	+1
Paper	+1	0	-1
Scissors	-1	+1	0



Revisiting the first game

The Keynesian beauty contest

- No rational player would play above $2/3 \cdot 1000 = 667$
- Hence, no rational player would play above $2/3 \cdot 667 = 445$
- What happens if we continue this logic?
- What would "rational" players play?

Dominance

- We say that strategy *A dominates* strategy *B* if playing *A* is always at least as good of an idea
- $u(A, O) \geq u(B, O)$ for all player 2 strategies *O*
- Second-order dominance: replace “is always a better idea” with “is always a better idea, if our opponent does not play dominated strategies”
- And so on...



Questions?

Dominance Example

		Player 2	
		C	D
Player 1	A	+3	+2
	B	+1	-1

Dominance Example

		Player 2	
		C	D
Player 1	A	+3	+2
	B	+1	-1

Dominance Example

		Player 2	
Player 1		C	D
	A	+3	+2
	B	+1	-1

Dominance Example

		Player 2	
Player 1		C	D
	A	+3	+2
	B	+1	-1



Equilibrium

A Nash equilibrium is
a set of strategies,
one for each player,
such that nobody has
an incentive to switch.

Nash equilibrium examples

- Keynesian beauty contest
 - All 0s is an equilibrium
- RPS?
 - $\frac{1}{3}$ rock, $\frac{1}{3}$ paper, $\frac{1}{3}$ scissors is an equilibrium

Solving the equilibrium for RPS

Player 1 probabilities: r, p, s

$$r + p + s = 1; r, p, s \geq 0$$

Player 2 probabilities: r', p', s'

$$r' + p' + s' = 1; r', p', s' \geq 0$$

We care about expected utility given each case of opponent's possible play

	Rock (r')	Paper (p')	Scissors (s')
Rock (r)	0	-1	+1
Paper (p)	+1	0	-1
Scissors (s)	-1	+1	0
EV Opponent			

Solve $p - s = s - r = r - p$ to get $r = p = s = 1/3$. This guarantees us at least 0 (yay I guess)

Why $p - s = s - r = r - p$?

- Suppose opponent plays the optimal r', p', s' counterstrategy against our r, p, s strategy
- Opponent minimizes the utility function amongst their three options:

$$\min_{r', p', s'} (p - s, s - r, r - p)$$

- We play the strategy that maximizes the utility function *even against the optimal counterstrategy*:

$$\max_{r, p, s} (\min_{r', p', s'} (p - s, s - r, r - p))$$



Questions?

Existence of Nash equilibria

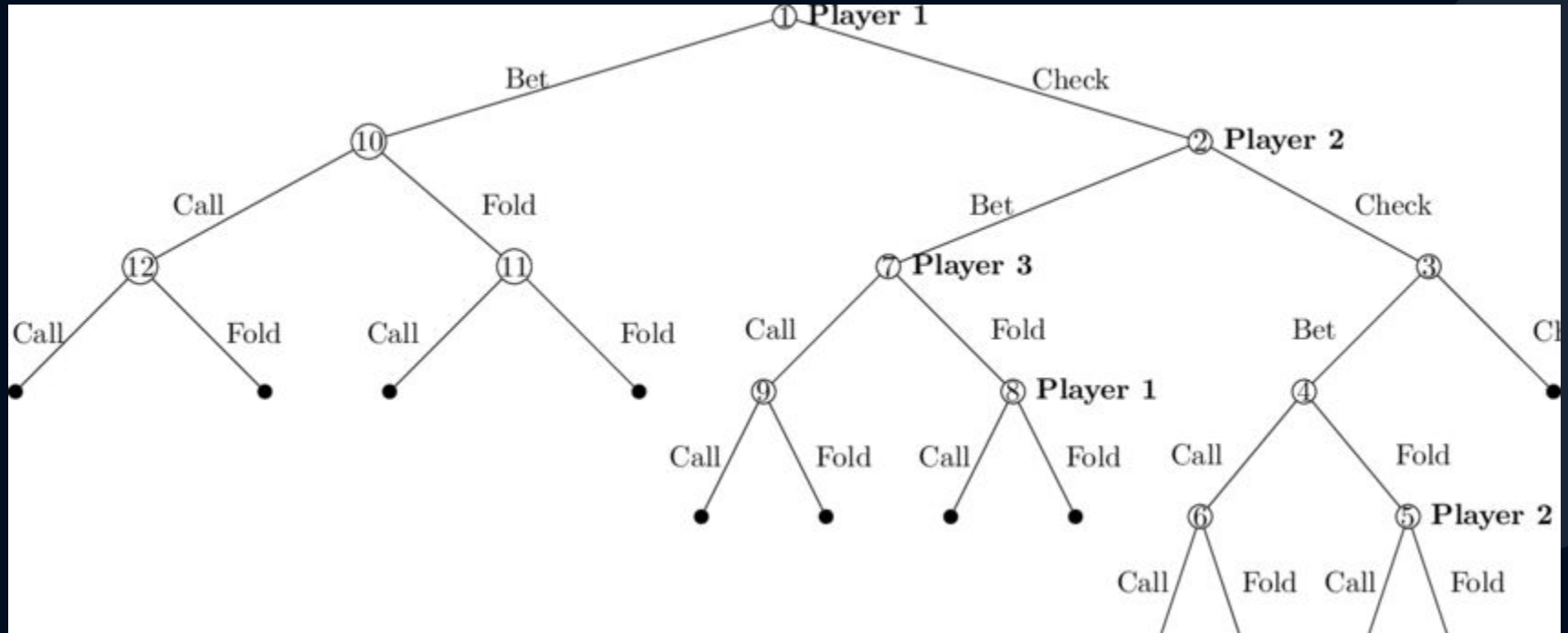
Theorem (Nash, 1951): Every finite game has a Nash equilibrium in mixed strategies.

Proof: Topology (uses Brouwer fixed-point theorem).

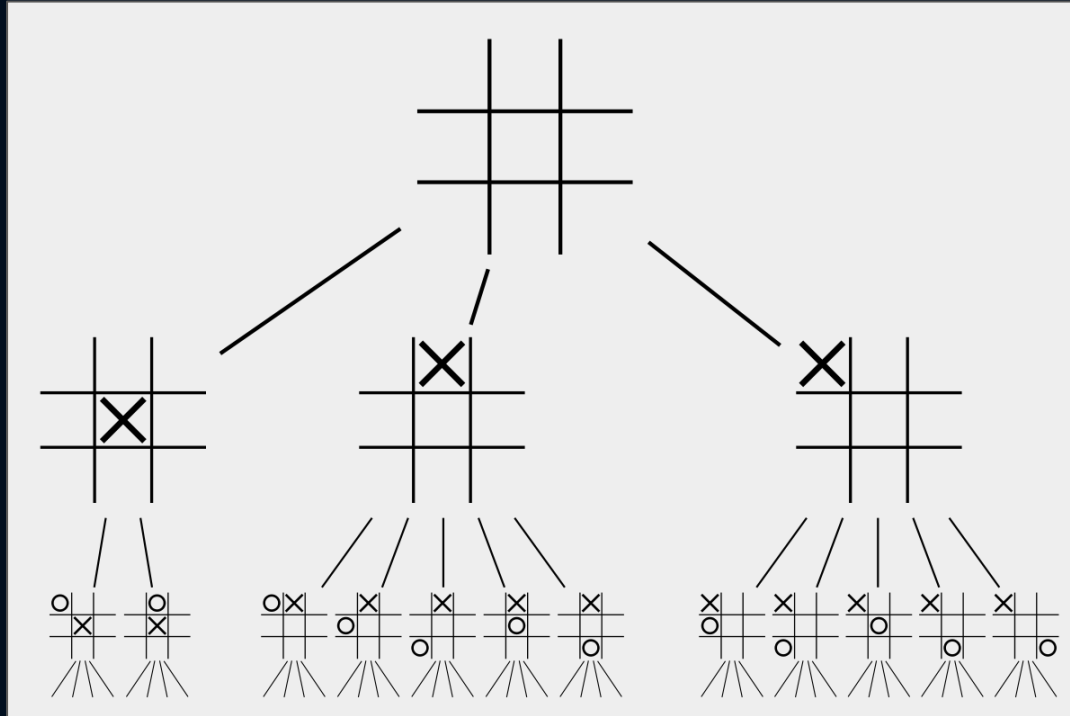


Applications to poker

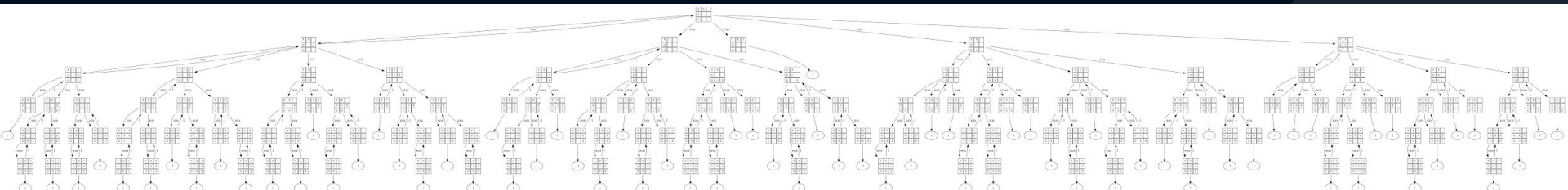
Game Tree



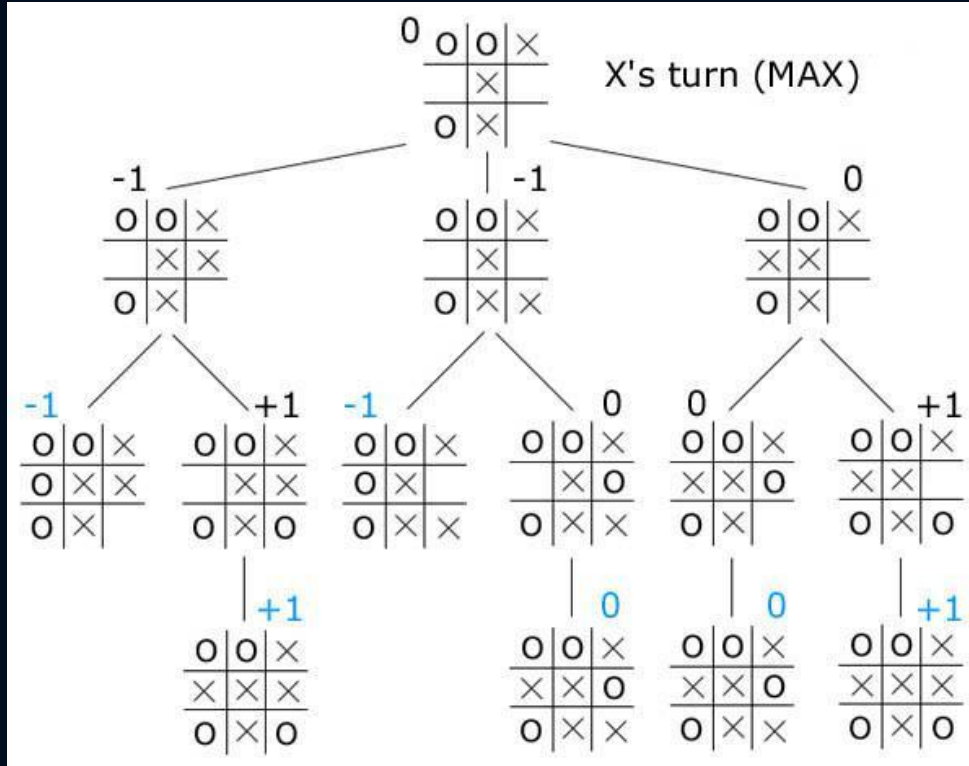
Extensive form games



Tic Tac Toe Full Extensive Form



Backwards induction



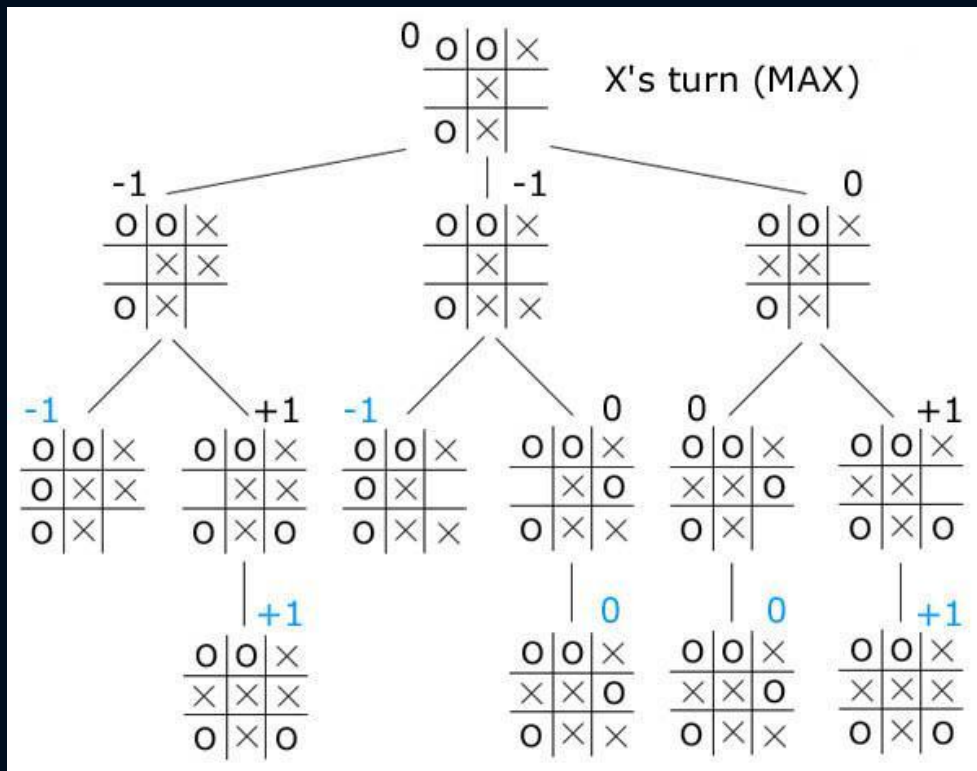
Black are the determined payouts based on players acting rationally

Blue are the leaves - the terminal states of the game

Follow the game tree up (backwards) to find the state's value

Minimax strategy

Backwards induction



Looking at the leaf states first, I see -1s in the left 2 branches, leaving their parents as -1.

The rightmost branch has payouts 0 and 1. Following this up the tree, we get 0 for the black value.

Finally, all together X should try to play for 0 (tie), and does so by putting x in left center.

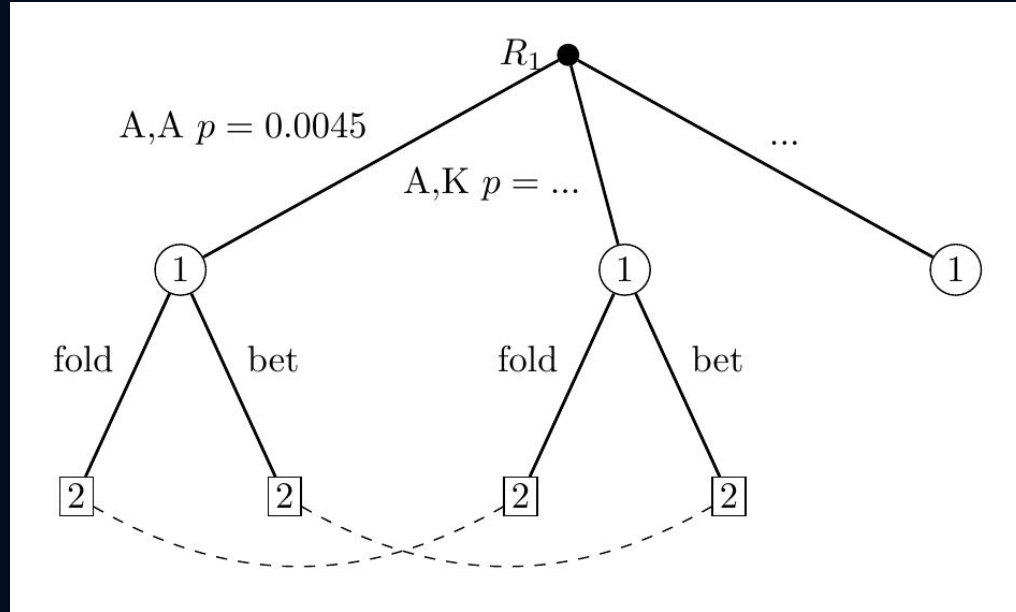
Conversion to normal form

- Every extensive form is also a normal form (matrix) game
- Mapping: each pure strategy (row/column) is a complete set of instructions for what to do at each node
- Example: in poker, each pure strategy would be a dictionary which maps the game state (info set) to a unique action

Implications

- Poker has a Nash equilibrium - it's a finite game that satisfies the conditions
- Can we use backwards induction to solve for it? Can we still use the same type of game tree?
- How is poker different from Chess? tic-tac-toe? RPS?
 - Size
 - Imperfect Information
 - Randomness

Adding in imperfect information and randomness



Imperfect Information and Randomness

- Imperfect: Don't know what node you're currently at since your opponent's hole cards are hidden (information sets)
- Randomness: Don't know where in the game tree you're moving next since future cards to be dealt are unknown
- Together, this leads to a really nice property of poker. The general game tree movements (bet/fold/call) are known, but the definite game state isn't. Imperfect information makes games fun!

Kuhn Poker

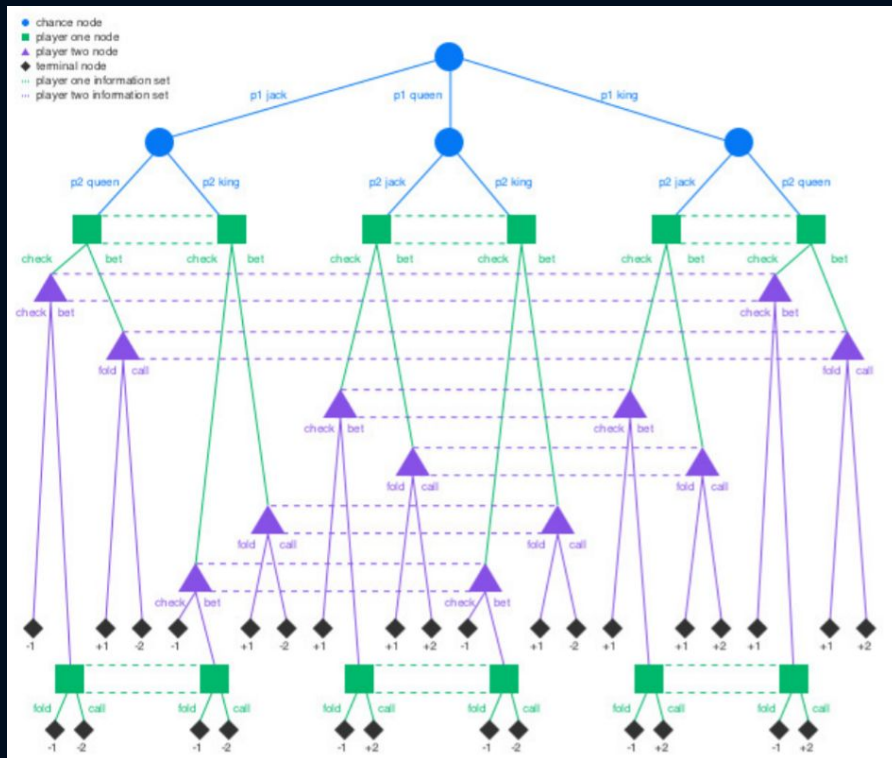
Kuhn poker is a simplified form of [poker](#) developed by [Harold W. Kuhn](#) as a simple model [zero-sum](#) two-player [imperfect-information](#) game, amenable to a complete [game-theoretic](#) analysis. In Kuhn poker, the deck includes only three [playing cards](#), for example a King, Queen, and Jack. One card is dealt to each player, which may place bets similarly to a standard poker. If both players bet or both players pass, the player with the higher card wins, otherwise, the betting player wins.

Game description [\[edit \]](#)

In [conventional poker terms](#), a game of Kuhn poker proceeds as follows:

- Each player [antes](#) 1.
- Each player is dealt one of the three cards, and the third is put aside unseen.
- Player one can [check](#) or [bet](#) 1.
 - If player one checks then player two can check or bet 1.
 - If player two checks there is a [showdown](#) for the pot of 2 (i.e. the higher card wins 1 from the other player).
 - If player two bets then player one can [fold](#) or [call](#).
 - If player one folds then player two takes the pot of 3 (i.e. winning 1 from player 1).
 - If player one calls there is a showdown for the pot of 4 (i.e. the higher card wins 2 from the other player).
 - If player one bets then player two can fold or call.
 - If player two folds then player one takes the pot of 3 (i.e. winning 1 from player 2).
 - If player two calls there is a showdown for the pot of 4 (i.e. the higher card wins 2 from the other player).

Kuhn Poker Game Tree



Poker Nash equilibrium and use

- Poker has a Nash equilibrium that has EV 0
 - Bluffing is included in this equilibrium
 - If you never 'lied' about your hand, your strategy would be **exploitable**
- The matrix is doubly exponential in size - incomputable
- Does “play good poker” reduce to “play the equilibrium strategy”?
- Exploitative strategies as alternatives



Applied game theory

Adverse selection

- Occurs anytime “buyer” and “seller” have asymmetric information
- Market for cars: suppose there are used cars with private value distributed uniformly between \$1000 and \$10000. What should the market price be?
- What would car owners do? Then what happens?
- Kerb bucket game

Sources of adverse selection in poker

- First action when betting (differential check to the preflop aggressor)
- Multiple bets in a row (what types of hands do this?)
- Being in the later stages of the round (why would they still be in the pot?)
- Tossing an Ace (what types of hands would do this?)

When I bet, am I
happy with my bet
*conditioned on my
opponent calling?*

Adverse selection and determinism

- Suppose I have the following deterministic betting strategy:
- All-in when I have a top $X\%$ hand
- How can my opponent exploit me?

Game Theory Takeaways

- Try to be one step ahead of your opponent (dominance)
- Equilibrium strategies can be good but hard to find
- Exploitative strategies can be very powerful - think about what you're trying to optimize for when making a pokerbot
- Be careful of adverse selection



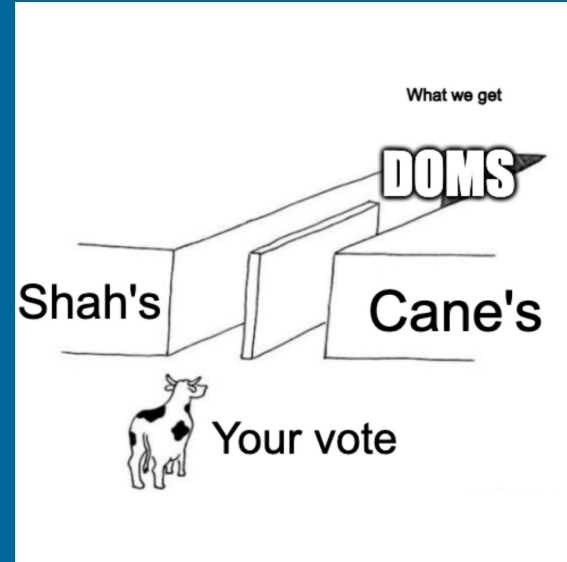
Questions?

During Lunch

Fill out this form

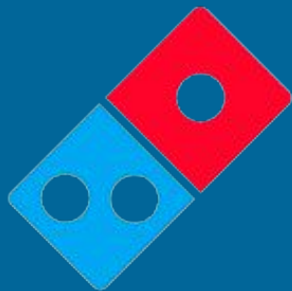
pkr.bot/lunch

Let your voice be heard in
our choice of lunch food



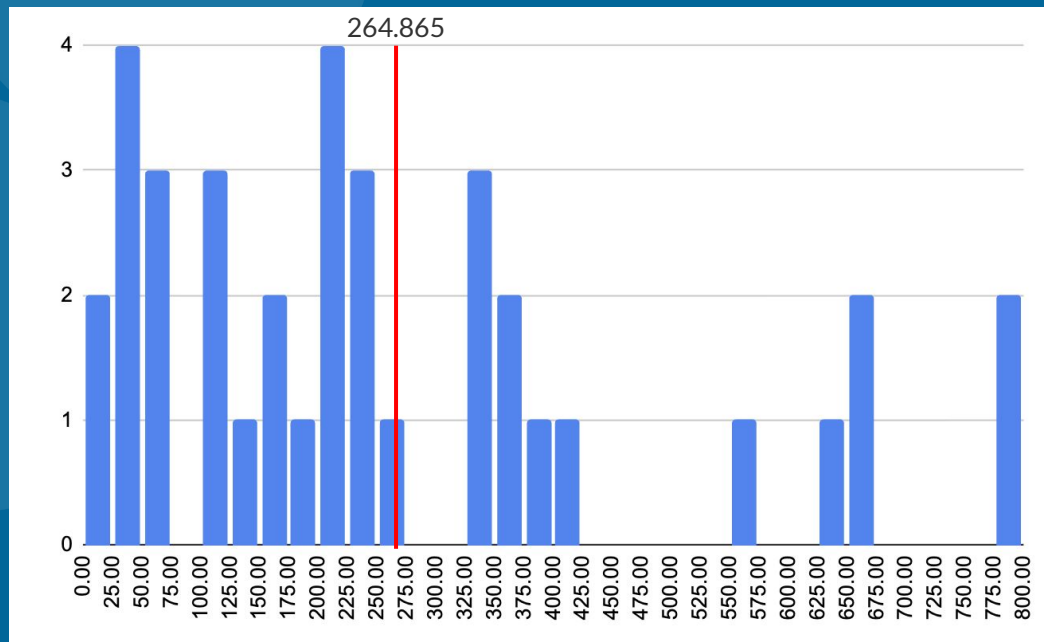
Lunch 🤗

Leave any type of feedback at pkr.bot/feedback!



Domino's

2/3 Contest Winner: vfang

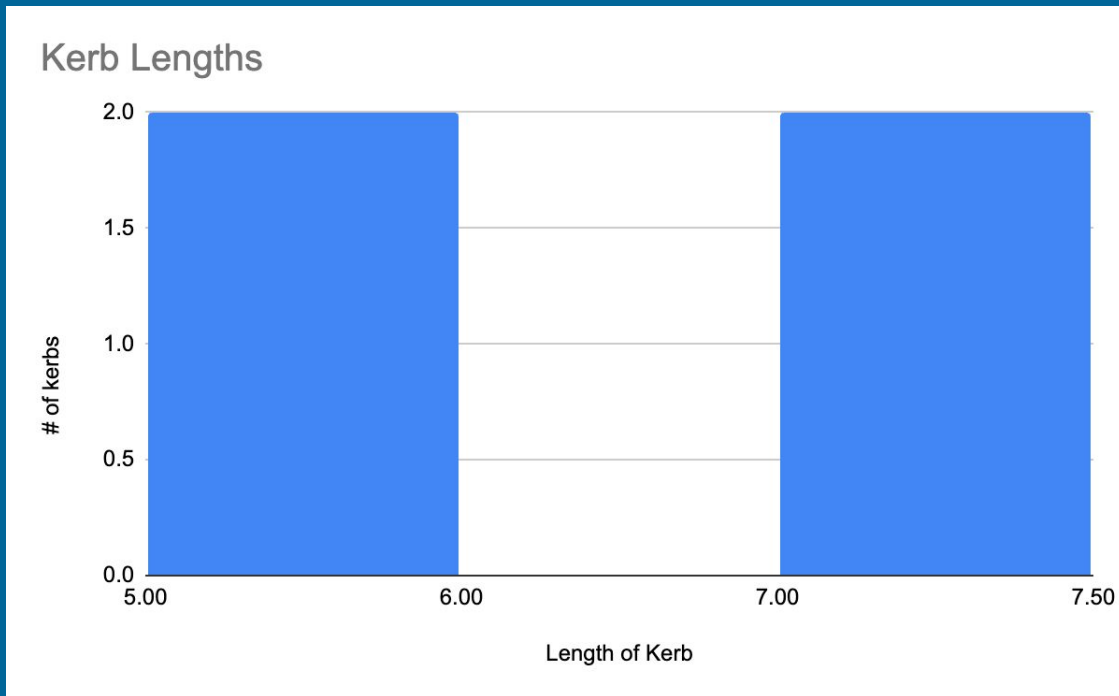


$\frac{2}{3}$ avg: 176.577

Guess: 185

Kerb Squared Contest Winners: tslu, zhank

- 4 submissions total
- Note: MIT limits kerb length to 8 characters
- flipped coin to decide!



Thanks for watching!

Information hub at pkr.bot/hub

Slides/notes will be posted on pkr.bot/resources

Make sure to check pkr.bot/piazza for updates

Lecture recordings at pkr.bot/panopto

Leave feedback at pkr.bot/feedback!